



Birla Institute of Technology & Science, Pilani

Work-Integrated Learning Programmes Division

MTech in Artificial Intelligence and Machine Learning

S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 1 – PS1 - Waste Collection - [Weightage 12%]

Read through this entire document very carefully before you start!

Smart Waste Collection Agent

Assume that you are the Municipal Commissioner of Bengaluru city. The city has implemented a Smart City initiative for efficient waste management and route optimization. Different garbage collection centers and waste processing units are located across the city and connected through road networks.

You are provided with a map containing important waste collection zones and roads connecting them. A Smart Waste Collection Robot Agent is assigned to transport waste efficiently between collection centers and processing plants.

The robot operates on limited battery power and fuel efficiency constraints. Therefore, it must determine the optimal route from a starting location to a target waste processing unit while minimizing the overall travel cost.

The city waste management network is represented as a weighted graph where:

- Vertices represent waste collection centers or processing units.
- Edges represent roads connecting those locations.
- Edge weights represent travel cost, distance, or fuel consumption.

The algorithm must accept the starting location and destination from the user and determine the most efficient route using intelligent search techniques.

The robot should avoid unnecessary travel and always attempt to minimize the total operational cost.

“To improve navigation efficiency, the Smart Waste Collection Robot Agent uses the intelligent path-finding algorithm IDA* to identify the least-cost route from the source node to the goal node. This algorithm uses heuristic estimates and path costs to efficiently explore the graph and determine the optimal solution.”.

Case 1:

Number of Locations: 6

Number of Roads: 7

Road Connections:

MG_Road Electronic_City 2

MG_Road Koramangala 4

Electronic_City Whitefield 2

Whitefield Yelahanka 4

Whitefield Jayanagar 6

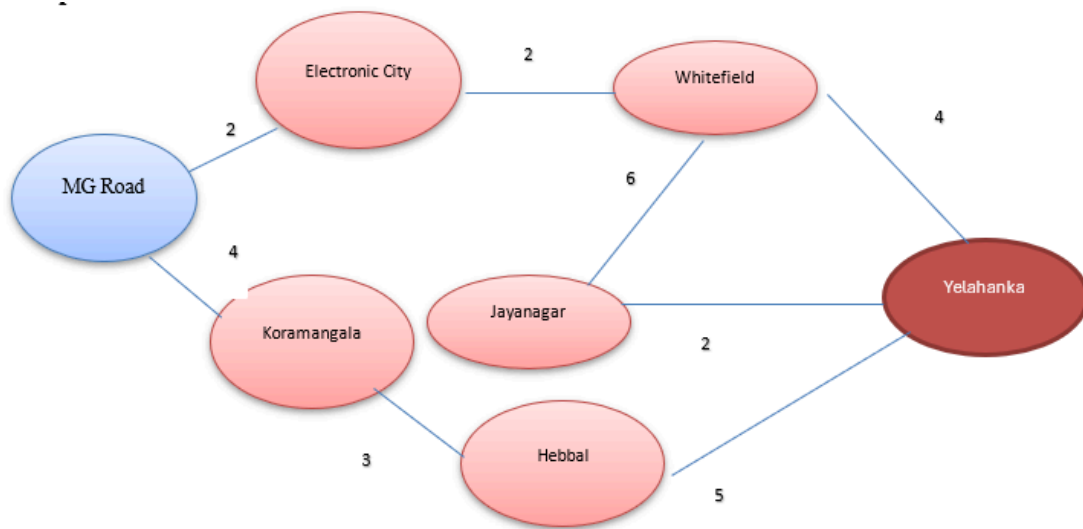
Jayanagar Yelahanka 2

Koramangala Hebbal 3

Hebbal Yelahanka 5

Source: MG_Road

Destination: Yelahanka



Case 2:

Number of Locations: 5

Number of Roads: 6

Road Connections:

A B 3

A C 2

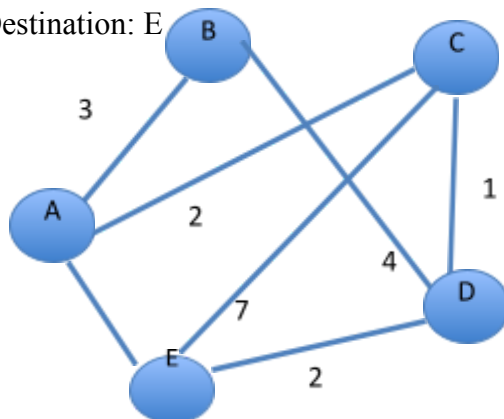
B D 4

C D 1

C E 7

D E 2

Source: A Destination: E



- Explain the heuristic used to solve this problem and justify your choice.
- Explain the cost function associated with the search in reaching the goal node.
- Identify and justify the informed search algorithm (IDA*) chosen for this problem.
- Implement the IDA* algorithm in Python.
- Print the following output from the algorithm:
 - The optimal path found from source to destination.
 - The total travel cost of the optimal route.
 - The number of nodes explored during the search.
 - The sequence of locations visited by the robot.

The assignment will be evaluated on the following points

1. PEAS analysis
2. Explanation of the heuristic and algorithm chosen
3. Implementation of the informed search in PYTHON
4. Print the details generated by the algorithm.

Deliverables:

- PDF document **designPSXX_<group id>.pdf** detailing your solution.
- **[Group id] _Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- .py file containing the python code. Create a single notebook. Do not fragment your code into multiple files
- Zip all of the above files including the design document in a folder with the name:
 - **[Group id] _A1_PSXX_XXXXXXXXXX.zip** and submit the zipped file.
 - Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.

- Ensure that all data structure insert and delete operations throw appropriate messages when their capacity is empty or full. Also ensure basic error handling is implemented.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions
- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not hard code any values into the code.
- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries in [this dedicated discussion section](#). Beware that only hints will be provided and queries asked **in other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

1. The strict deadline for submission of the assignment is **8th June 2026 11:55 PM IST (Monday)**.
2. The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
3. Late submissions will be evaluated as below. Submissions made between:
 - a. 8th June 2026 11:56PM to 9th June 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - b. 9th June 2026 11:56PM to 10th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6marks.
 - c. Beyond 10th June 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id] _A1_PSXX_XXXXXXXXXX.zip and submit the zipped file.

- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means like email etc. will not be graded.

Evaluation

- The assignment carries **12 Marks**.
- Grading will depend on:
- Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.